Disas AI Engine

GNANI.AI HACKATHON SHOWCASE

Voice AI & Spatial Intelligence Gateway for Disaster Response

> **Mission:** Bridging Satellite Predictive AI and Last-Mile Regional Voice Communication in High-Noise Emergency Environments.

1. OPERATIONAL PROBLEM & EXECUTIVE SUMMARY

⚠️ Inbound Distress Noise Bottleneck

Satellite phone & PSTN calls from storm-affected citizens suffer severe acoustic distortion (torrential rain, rushing water, codec compression), rendering distress calls unintelligible to human dispatchers.

📡 Outbound Last-Mile Language Gap

Traditional smartphone app alerts fail during power grid blackouts and cellular data collapse. Furthermore, text broadcasts ignore non-literate populations and regional language barriers in rural Indian sectors.

2. ARCHITECTURE: DISASTER-IN & DISASTER-OUT CLOSED LOOP



3. TECHNICAL PHASE EXECUTION

Phase 1: Disaster-In Intake

Inbound Acoustic AI

- **Signal Denoising:** Ingests distorted sat-phone audio through a **Butterworth Bandpass Filter (250Hz–3400Hz)** paired with a Neural Speech Enhancer.
- **Gnani Prisma STT:** Transcribes cleaned audio streams into localized text across low-resource Indian dialects with high precision.
- **LLM Dispatch Vectoring:** Extracts critical triage metrics (landmarks, stranded headcount, flood depth, medical urgency) to auto-generate rescue tickets.

Phase 2: Disaster-Out Warning

Outbound Spatial AI

- **PyTorch Spatial Engine:** Evaluates SAR/Optical satellite rasters using a **U-Net** (susceptibility) and **ConvLSTM** (6–12h rainfall forecast).
- **GPS Language Mapping:** Automatically maps coordinates (e.g., Kochi ➔ Malayalam, Kolkata ➔ Bengali) without manual intervention.
- **Gnani Vachana TTS:** Synthesizes regional voice warnings streamed directly over low-bandwidth IVR networks to basic $10 feature phones.

4. INTEGRATED TECH STACK

Layer	Technology / Module
Voice AI Engine	Gnani.ai Prisma STT & Vachana TTS REST APIs
Signal Processing	SciPy Butterworth Bandpass Filter & DSP Enhancer
Spatial ML Model	PyTorch U-Net (Inundation) & ConvLSTM (Rainfall)
Intelligence & UI	LLM Dispatch Parsing & Streamlit Microservices

5. COMPETITIVE ADVANTAGE

1. **Hardware-Grade Noise Resilience:** Filters sat-phone audio *before* running speech recognition, avoiding STT breakdown during storms.
2. **Deep Gnani Core Integration:** Uses Gnani.ai as the heroic bridge connecting deep learning spatial predictions to human action.
3. **Zero-Internet Last-Mile Reach:** Delivers voice calls directly to basic feature phones without requiring smartphones or cellular data.